```
#(1 2 3 4) collect: [ :each | each * 2 ]    ⟶    #( 2 4 6 8 )
#(1 2 3 4)
    inject: 0
    into: [ :each :result | each + result ]    ⟶    10
```
1

```
" testing "
#( 2 4 ) anySatisfy: [ :each | each odd ]    ⟶    false
#( 2 4 ) allSatisfy: [ :each | each even ]    ⟶    true
```
2

```
" finding "
'abcdef' includes: $e    ⟶    true
'abcdef' contains: [ :each | each isUppercase ]    ⟶    false
'abcdef'
    detect: [ :each | each isVowel ]
    ifNone: [ $u ]    ⟶    $a
```
3

```
" String – a collection of characters "
string := 'abc'.
string := string , 'DEF'    ⟶    'abcDEF'
string beginsWith: 'abc'    ⟶    true
string endsWith: 'abc'    ⟶    false
string includesSubString: 'cD'    ⟶    true
string asLowercase    ⟶    'abcdef
string asUppercase    ⟶    'ABCDEF'
```
4

```
" OrderedCollection – an ordered collection of objects "
ordered := OrderedCollection new.
ordered addLast: 'world'.
ordered addFirst: 'hello'.
ordered size    ⟶    2
ordered at: 2    ⟶    'world'
ordered removeLast    ⟶    'world'
ordered removeFirst    ⟶    'hello'
ordered isEmpty    ⟶    true
```
5

```
" Set – an unordered collection of objects without duplicates "
set := Set new.
set add 'hello'; add: 'hello'.
set size    ⟶    1
```
6

```
" Bag – an unordered collection of objects with duplicates "
bag := Bag new.
bag add: 'this'; add: 'that'; add: 'that'.
bag occurrencesOf: 'that'    ⟶    2
bag remove: 'that'.
bag occurrencesOf: 'that'    ⟶    1
```
7

```
" Dictionary – associates unique keys with objects "
dictionary := Dictionary new.
dictionary at: 'smalltalk' put: 80.
dictionary at: 'smalltalk'    ⟶    80
dictionary at: 'squeak' ifAbsent: [ 82 ]    ⟶    82
dictionary removeKey: 'smalltalk'.
dictionary isEmpty    ⟶    true
```
8

Streams 9

```
" ReadStream – to read a sequence of objects from a collection "
```
10

```
stream := 'Hello World' readStream.
stream next    ⟶    $H
stream upTo: $o    ⟶    'ell'
stream skip: 2.
stream peek    ⟶    $o
stream upToEnd    ⟶    'orld'
```
11

```
" WriteStream – to write a sequence of objects to a collection "
stream := WriteStream on: Array new.
stream nextPut: 'Hello'.
stream nextPutAll: #( 1 2 3 ).
stream contents    ⟶    #( 'Hello' 1 2 3 )
```
12

File Streams 13

```
fileStream := FileDirectory default newFileNamed: 'tmp.txt'.
fileStream nextPutAll: 'my cool stuff'.
fileStream close.
```
14

```
fileStream := FileDirectory default oldFileNamed: 'tmp.txt'.
fileStream contents    ⟶    'my cool stuff'
```
15

Method Definition 16

```
messageSelectorAndArgumentNames
    "comment stating purpose of message"
    | temporary variable names |
    statements
```
17

Class Definition 18

```
Object subclass: #NameOfSubclass
    instanceVariableNames: 'instVar1 instVar2'
    classVariableNames: ''
    poolDictionaries: ''
    category: 'Category-Name'
```
19

References 20

1. Andrew Black, Stéphane Ducasse, Oscar Nierstrasz and Damien Pollet, *Squeak by Example*, Square Bracket Associates, 2007, `squeakbyexample.org`.
2. Chris Rathman, *Terse guide to Squeak*, `wiki.squeak.org/squeak/5699`.
3. *Smalltalk*, Wikipedia, the free encyclopedia, `en.wikipedia.org/wiki/Smalltalk`.

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University of Bern

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1. The Environment 25

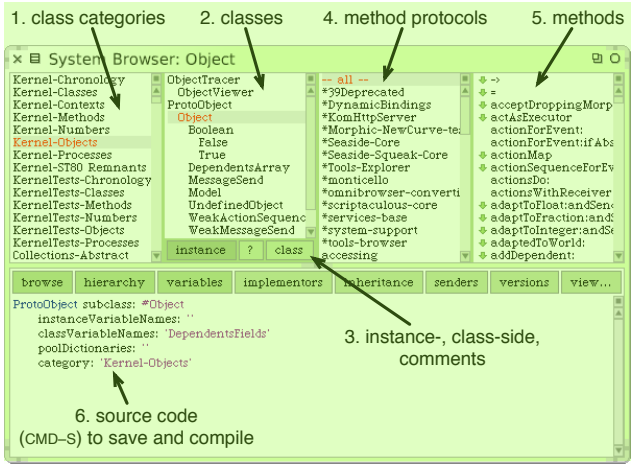

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Figure 1: The Smalltalk Code Browser 27

- Do it (CMD–D): Evaluate selected code. 28
- Print it (CMD–P): Display the result of evaluating selected code.
- Debug it: Evaluate selected code step-by-step with the integrated debugger.
- Inspect it (CMD–I): Show an *object inspector* on the result of evaluating selected code.
- Explore it (CMD–SHIFT–I): Show an *object explorer* on the result of evaluating selected code.

2. The Language 1

- Everything is an object. 2
- Everything happens by sending messages.
- Single inheritance.
- Methods are public.
- Instance variables are private to objects.

Keywords 3

- self, the receiver. 4
- super, the receiver, method lookup starts in superclass.
- nil, the unique instance of the class UndefinedObject.
- true, the unique instance of the class True.
- false, the unique instance of the class False.
- thisContext, the current execution context.

Literals 5

- Integer 6
  123
  2r1111011 (123 in binary)
  16r7B (123 in hexadecimal)
- Float
  123.4
  1.23e−4
- Character
  $a
- String
  'abc'
- Symbol
  #abc
- Array
  #(123 123.4 $a 'abc' #abc)

Message Sends 7

1. *Unary messages* take no argument. 8
   1 factorial sends the message factorial to the object 1.
2. *Binary messages* take exactly one argument. 9
   3 + 4 sends message + with argument 4 to the object 3.
   #answer −> 42 sends −> with argument 42 to #answer.
   Binary selectors are built from one or more of the characters +, −, ⋆, =, <, >, ...
3. *Keyword messages* take one or more arguments. 10
   2 raisedTo: 6 modulo: 10 sends the message named 11
   raisedTo:modulo: with arguments 6 and 10 to the object 2.

Unary messages are sent first, then binary messages and 12
finally keyword messages:

2 raisedTo: 1 + 3 factorial ⟶ 128 13

Messages are sent left to right. Use parentheses to change 14
the order:

1 + 2 ⋆ 3 ⟶ 9 15
1 + (2 ⋆ 3) ⟶ 7

Syntax 16

- Comments 17
  "Comments are enclosed in double quotes"
- Temporary Variables
  | var |
  | var1 var2 |
- Assignment
  var := aStatement
  var1 := var2 := aStatement
- Statements
  aStatement1. aStatement2
  aStatement1. aStatement2. aStatement3
- Messages
  receiver message (unary message)
  receiver + argument (binary message)
  receiver message: argument (keyword message)
  receiver message: argument1 with: argument2
- Cascade
  receiver message1; message2
  receiver message1; message2: arg2; message3: arg3
- Blocks
  [aStatement1. aStatement2]
  [:argument1 | aStatement1. aStatement2]
  [:argument1 :argument2 | | temp1 temp2 | aStatement1]
- Return Statement
  ^ aStatement

3. Standard Classes 18

Logical expressions 19

true not ⟶ false 20
1 = 2 or: [2 = 1] ⟶ false
1 < 2 and: [2 > 1] ⟶ true

Conditionals 21

1 = 2 ifTrue: [Transcript show: '1 is equal to 2']. 22
1 = 2 ifFalse: [Transcript show: '1 isn''t equal to 2'].

100 factorial / 99 factorial = 100 23
ifTrue: [Transcript show: 'condition evaluated to true']
ifFalse: [Beeper beep].

Loops 24

" conditional iteration " 25
[Sensor anyButtonPressed]
whileFalse: [*"wait"*].

pen := Pen newOnForm: Display. 26
pen place: Sensor cursorPoint.
[Sensor anyButtonPressed]
whileTrue: [pen goto: Sensor cursorPoint].

" fixed iteration " 27
180 timesRepeat: [
pen turn: 88.
pen go: 250].

1 to: 100 do: [:index | 28
pen go: index ⋆ 4.
bic turn: 89].

" infinite loop (press CMD+. to break) " 29
[pen goto: Sensor cursorPoint] repeat.

Blocks (anonymous functions) 30

" evaluation " 31
[1 + 2] value ⟶ 3
[:x | x + 2] value: 1 ⟶ 3
[:x :y | x + y] value: 1 value: 2 ⟶ 3

" processes " 32
[(Delay forDuration: 5 seconds) wait.
Transcript show: 'done'] fork ⟶ aProcess

Collections 33

" iterating " 34
'abc' do: [:each | Transcript show: each].
'abc'
do: [:each | Transcript show: each]
separatedBy: [Transcript cr].

" transforming " 35
#(1 2 3 4) select: [:each | each even] ⟶ #(2 4)
#(1 2 3 4) reject: [:each | each = 2] ⟶ #(1 3 4)